

LabUMat 2

Service training

Introducing the LabUMat 2



LabUMat 2 parameters



LabStrip U11 Plus GL strip parameters

BILIRUBIN	 neg.	 +	 ++	 +++			
UROBILINOGEN	 norm.	 2(35)	 4(70)	 8(140)	 12(200)mg/dl(μmol/l)		
KETONES	 neg.	 +	 ++	 +++			
ASCORBIC ACID	 neg.	 +	 ++				
GLUCOSE	 norm.	 50(2.8)	 150(8)	 500(28)	 ≥1000(56)mg/dl(mmol/l)		
PROTEIN	 neg.	 30	 100	 ≥500mg/dl			
ERYTHROCYTES	 neg.	 +	 ++	 +++	 ca.5-10	 ca.50	 ca.300Erys./μl
pH	 5	 6	 7	 8	 9		
NITRITE	 neg.	 ca.25	 ca.75	 ca.500 Leukos./μl			
LEUKOCYTES	 neg.	 ca.25	 ca.75	 ca.500 Leukos./μl			
COMPENSATION	 neg.	 ca.25	 ca.75	 ca.500 Leukos./μl			
SPECIFIC GRAVITY	 1.000	 1.005	 1.010	 1.015	 1.020	 1.025	 1.030



pink/red colour = positive
Vertärbungen rosa/rot = positiv
Cualquier color rosado = positivo



PMC parameters

TURBIDITY

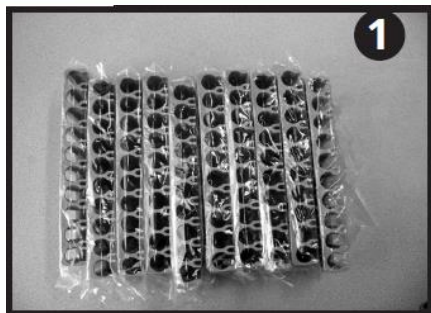
COLOR

SPECIFIC GRAVITY



All together 13 parameters

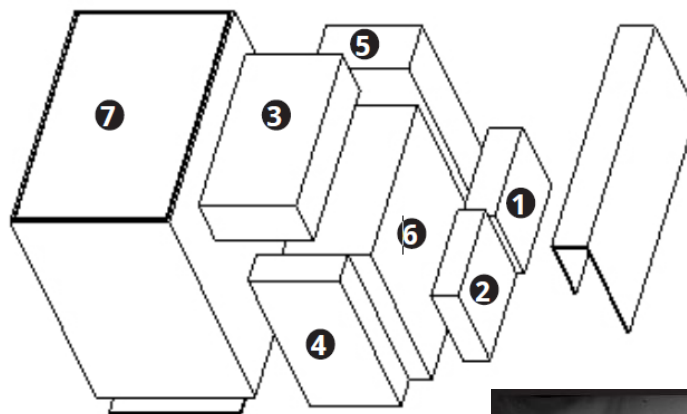
LabUMat 2 packaging content



Ten (10) racks with a capacity of ten (10) test tubes each [UAZ-4439-3]



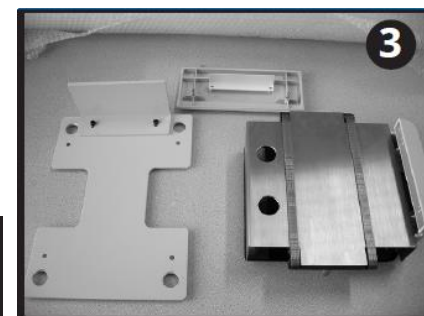
- Power cable [352HAL01]
- Microfiber cleaning wipe [90494TORK]
- Touchscreen stylus [61100001]
- Roll of thermal printer paper [612LT245]
- LabUMat 2 user's manual on CD [UPA-9250-1]
- Comb conveyor [UPA-033511-1]
- Pipetting stage [UPA-251159-2]
- Leveling disks [URM-980133-1]
- Water filter [55FV2B10]
- Cable ties [3C100002]
- Serial UriSed-to-LabUMat interface cable [ALT-466001-1]
- 12-milliliter graduated sample test tube [9E110X17]
- Drainboard [UPA-051177-2]



LabUMat 2 rack conveyor [UPA-441400-2]



- One hundred (100) 12-milliliter test tubes [9E110X17]
- Test tube stopper caps [9E065793]

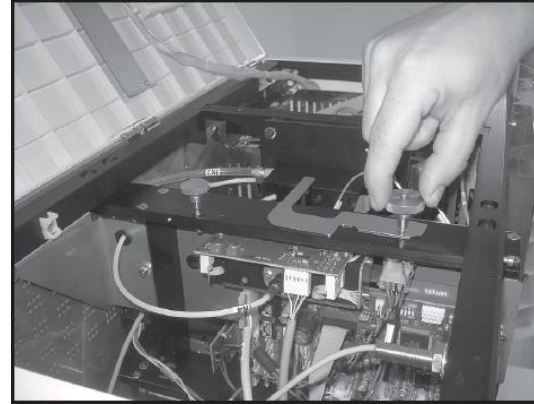
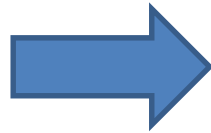
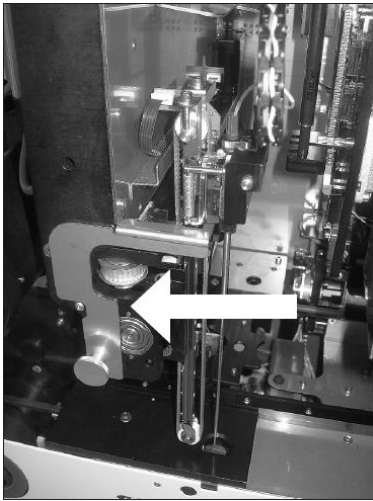


- LabUMat 2-to-UriSed rack conveyor connection element [UPA-440600-1]
- LabUMat 2-to-UriSed spreader plate [UPA-441900-2]
- Rack conveyor side panel [UPA-441414-1]

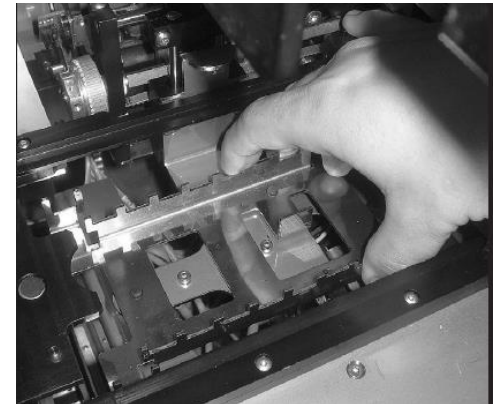
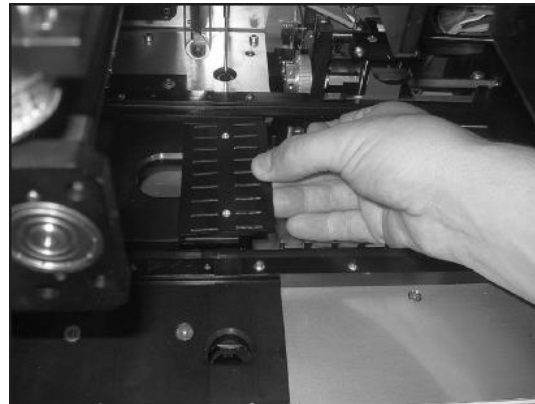
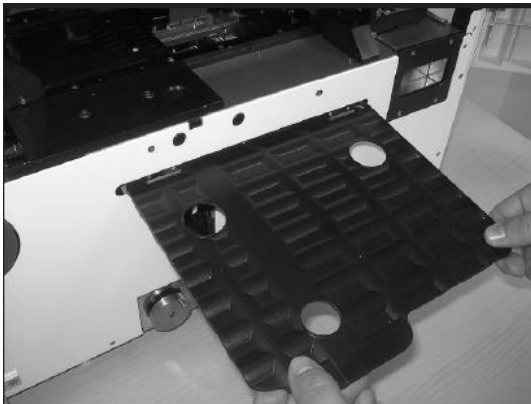


- 5-liter waste liquid tank [URS-447700-1]
- 5-liter distilled water tank [URS-447800-1]
- waste line tube (DIR1) [URS-1722-1]
- gravity line tube (GRA1) [URS-1721-1]
- distilled water tube (W3) [URS-1723-1]
- liquid tank stand [URS-3004-1]

LabUMat 2 installation I.

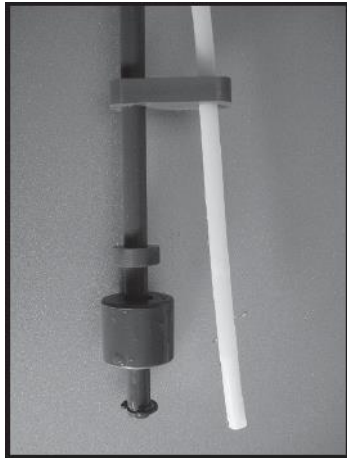


Remove the pipetting probe retainer plate then fix it on the top of the instrument.



Push the linear robot all the way to the back of the device and insert the drainboard, the pipetting stage, and the comb conveyor.

LabUMat 2 installation II.

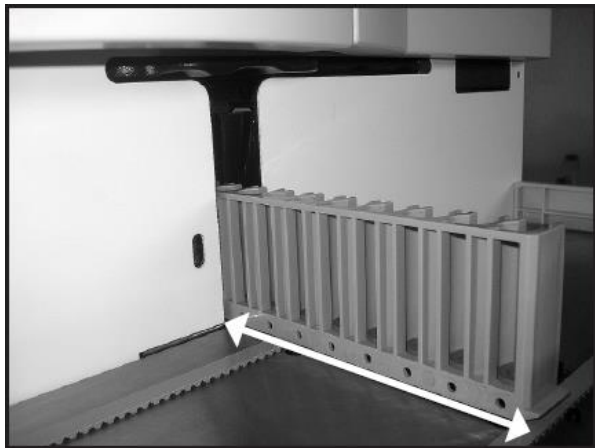


Push the tube into the tank so that the 'Push until' label on the tube is exactly above the flexible tube guide on the cap of the tank.

Make sure that the end of the tube extends beyond the float switch rod



Make sure that when the liquid tanks are in their final location, the gravity line tube is dropping all the way to the Waste tank. Avoid having the gravity tube double back upon itself.



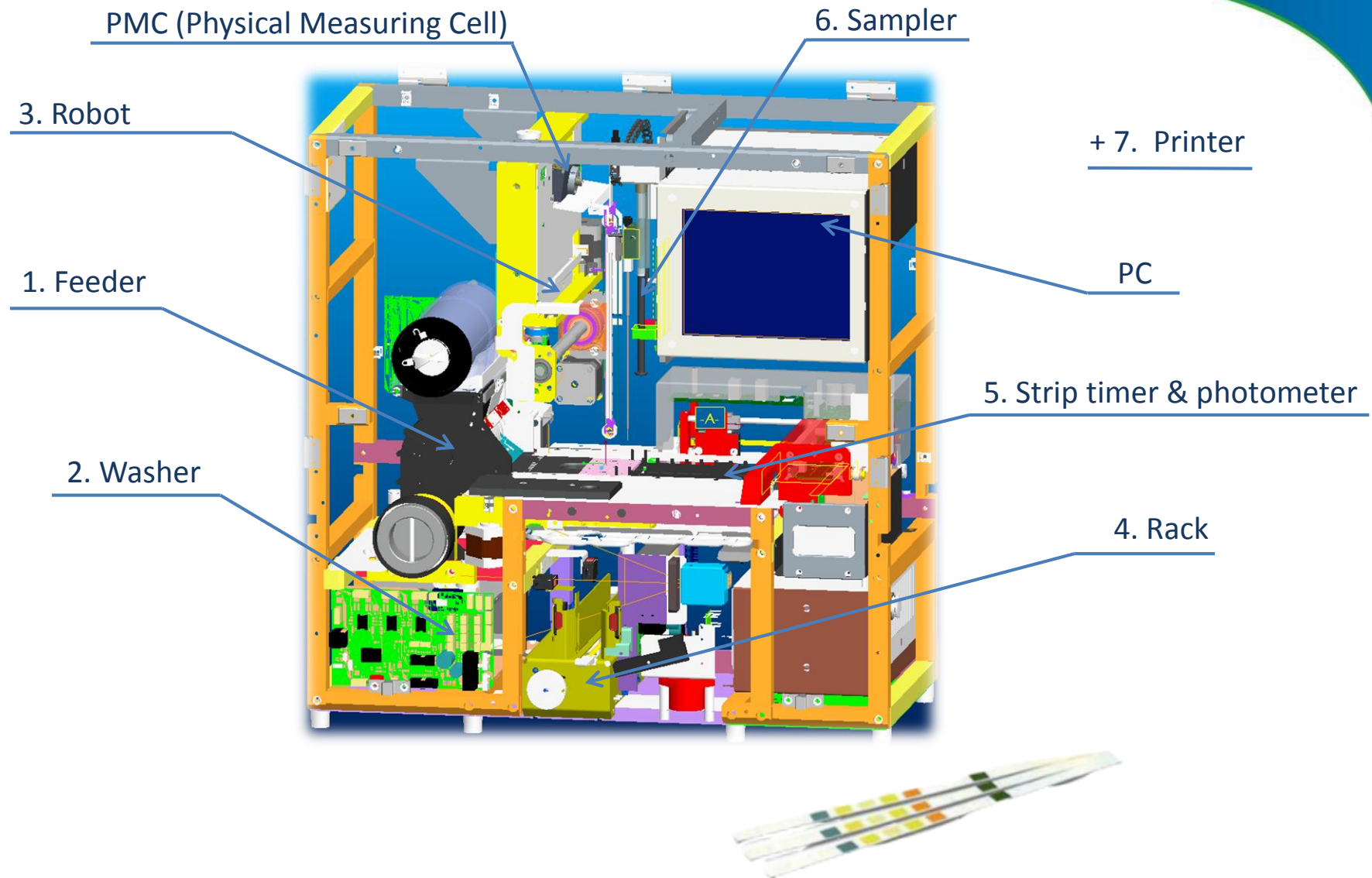
Place a test tube rack on the conveyor belts in the position where it is drawn into the device during normal operation.

Make sure that the rack conveyor is flush with the front casing panel of the device.

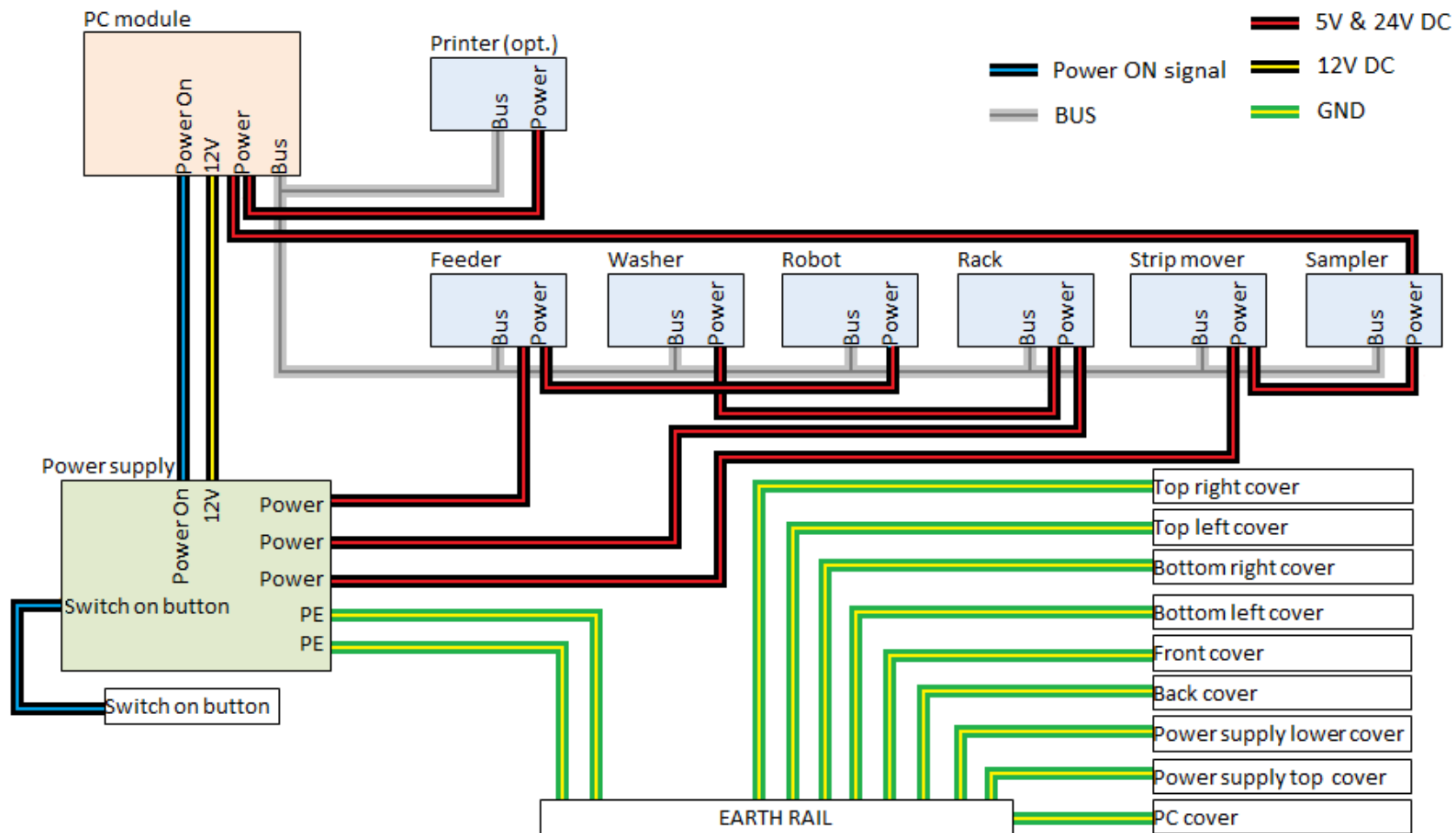
The test tube rack should be able to be drawn into and pushed out of the device without any tilting or jerking.



Overview of the main modules

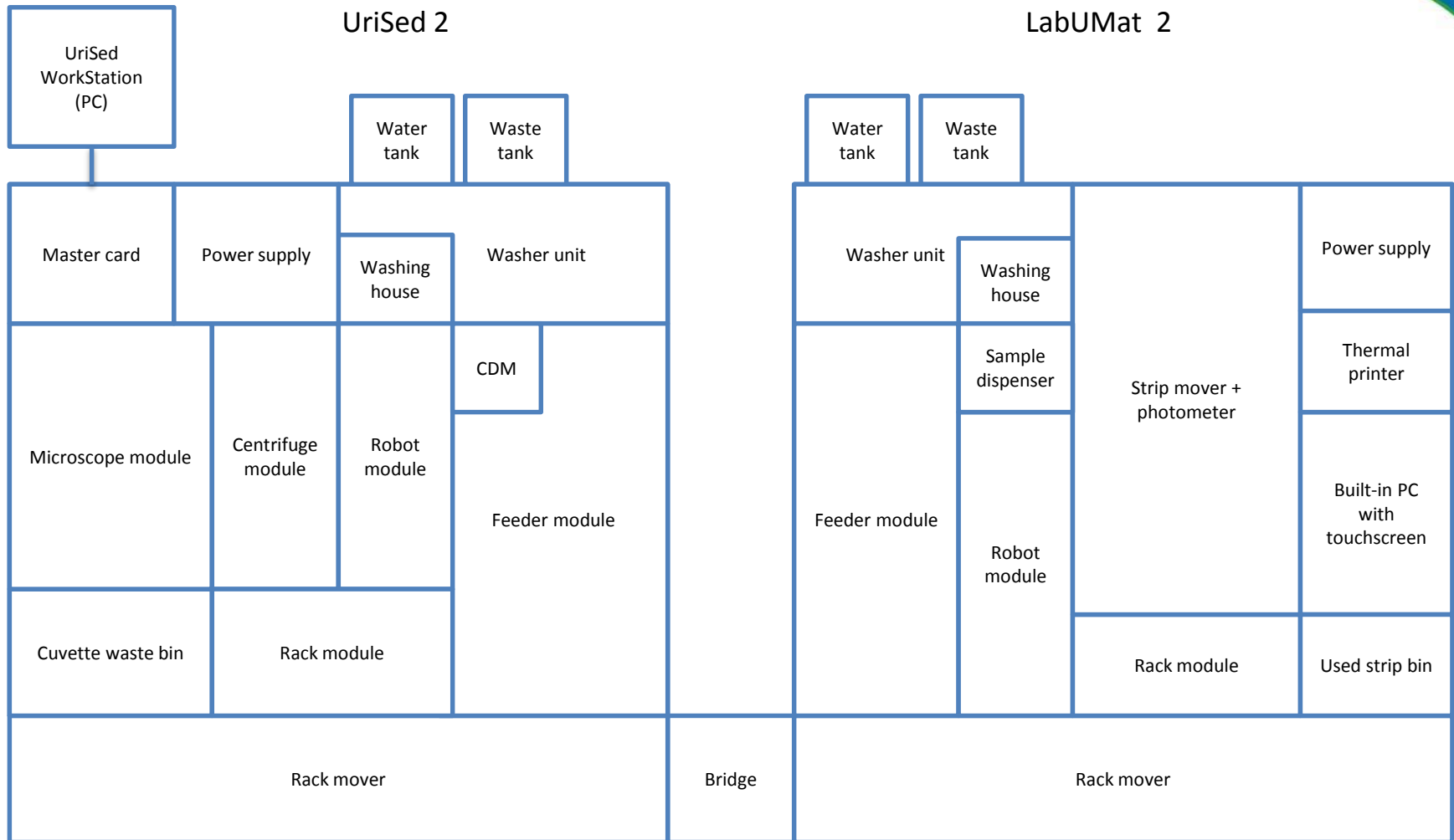


Block diagram

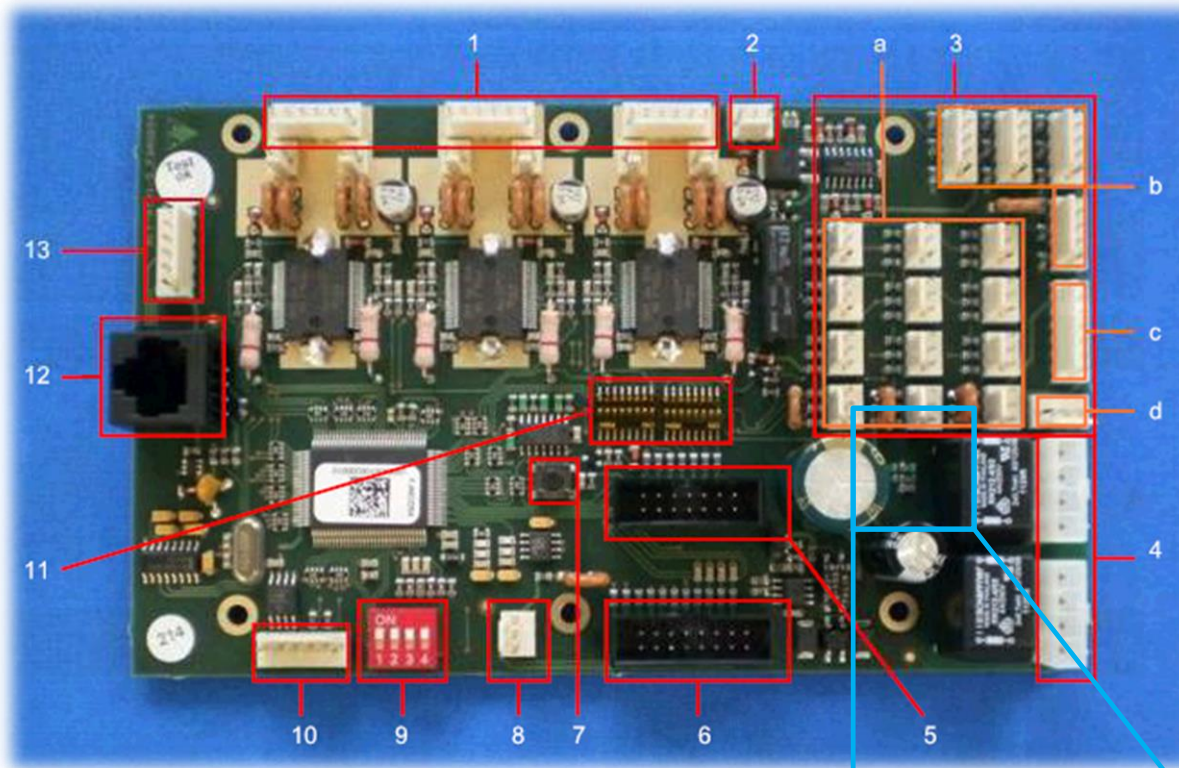




Modular structure of the connected systems



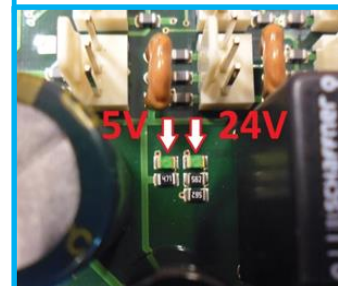
Motor controller board



- 1 - motor drivers
- 2 - n/a
- 3 - sensors
- 4 - power supply (two in parallel)
- 5 - bus communication (RS-485)
- 6 - internal communication (I2C)
- 7 - reset
- 8 - n/a
- 9 - dip switches for card addresses
- 10 - firmware programming
- 11 - jumpers
- 12 - internal communication (real RS-232)
- 13 - internal communication

- a - 5V
- b - 24V (parallel to a)
- c - 5V / 24V (parallel to a)
- d - 5V (parallel to a)

Power indicator LEDs:





Motor controller board replacement

1. Take a module specific motor controller PCB.
2. Before replacement please check the jumpers and the addressing of the DIP switch regarding the table on „Circuit board DIP switch settings” slide.
3. Replace the PCB
4. Start the service software and upgrade the card's low level software
5. Set the instrument serial number
6. Use position setup if neccessary (pipetting module, strip timer and photometer module)
7. Start User software



Circuit board DIP switch settings

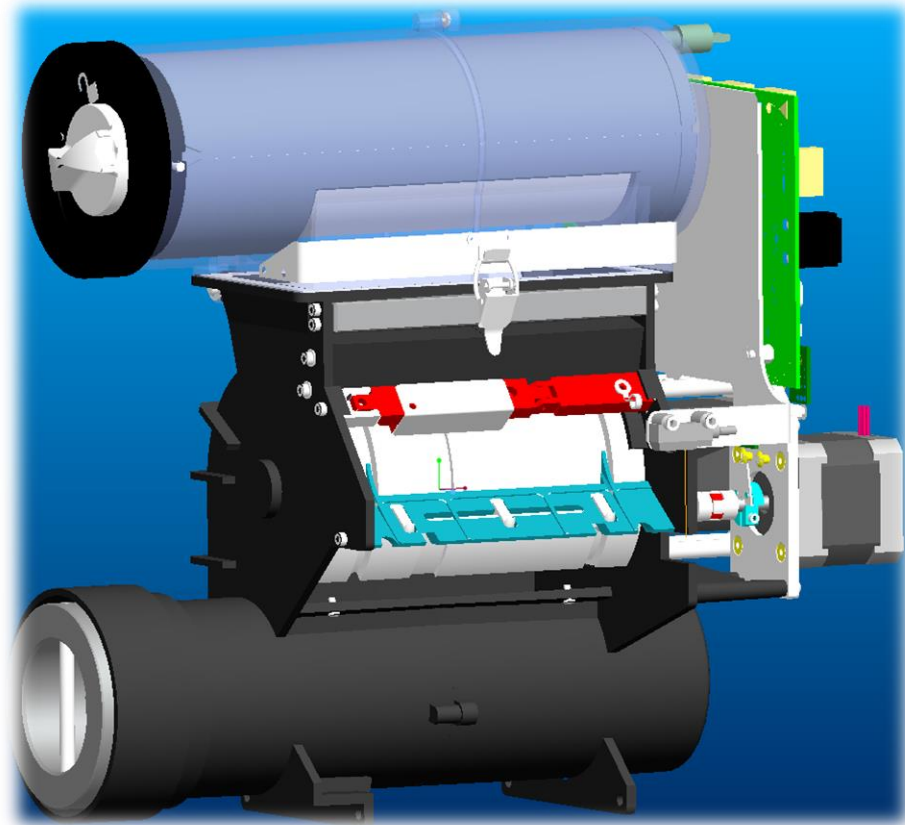
Module	Address	DIP switch	DIP „ON”	JUMPER* PCB: FJMOD55	SW3 „ON”	SW2 „ON”
Strip feeder	1		2,3,4		-	-
Washer	2		1,3,4		-	5
Robot	3		3,4		2	4,8
Rack	4		1,2,4		2	-
Strip timer and photometer	5		2,4	All solder jumpers are free	-	-
Sampler	6		1,4		-	4,5,8
Printer	7	n/a	n/a	n/a	n/a	n/a

*JUMPER: SW3 - see the left side of figure, SW2 – see the right side of figure

1. Strip feeder module

Main components

- strip loader cylinder
- feeder drum
- flipped strip sensor
- strip turner wheels
- strip pusher with strip clamps
- unused strip bin



Module description:

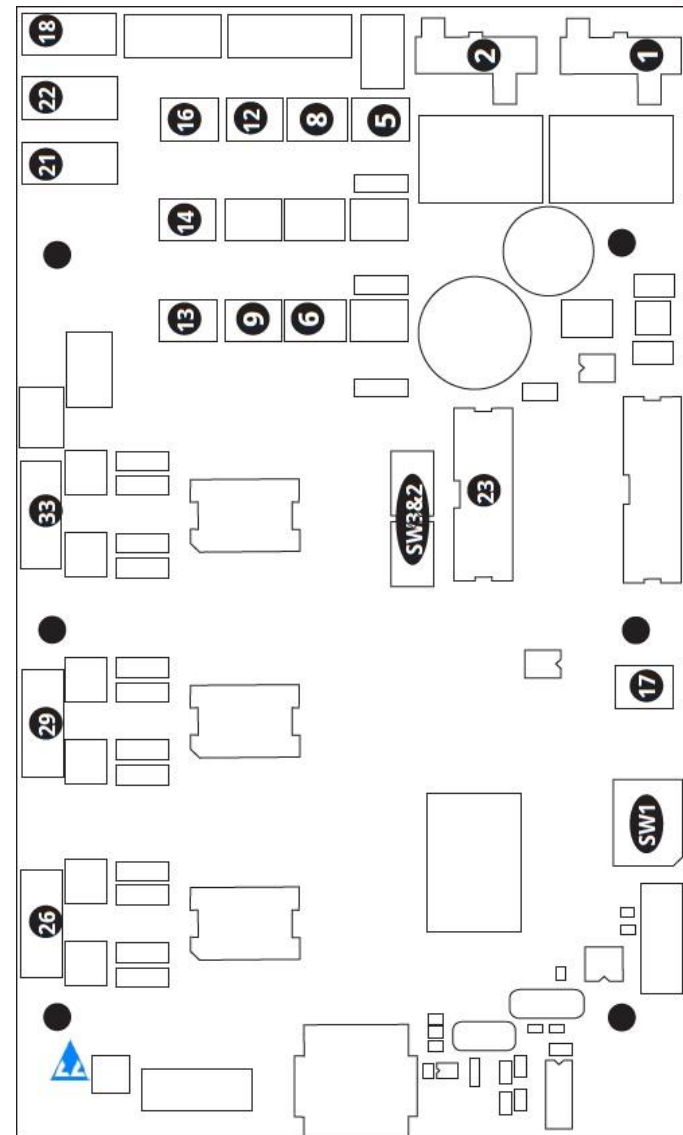
The strip feeder is responsible for dispensing and forwarding the strips one by one from the strip loader to the pipetting position.



1. Strip feeder module

PCB layout:

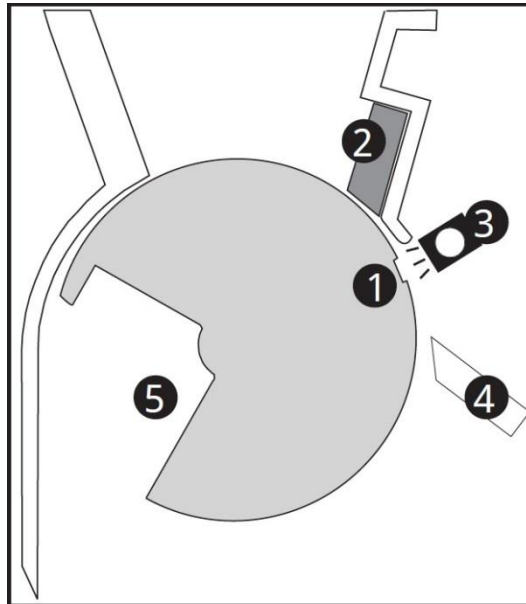
1. Power connector for the power supply
2. Power connector from PCB No. 3
5. Home position sensor connector
6. Strip turner home position sensor connector
8. Feeder motor „drop“ position sensor connector
9. Strip pusher „home“ position sensor connector
12. Feeder motor „upper empty“ position sensor
13. ~~Strip pusher pipetting position sensor connector~~
14. Strip feeding tube micro-switch connector
16. Feeder motor „lower empty“ position connector
17. Flipped strip sensor connector
18. Pipetting stage presence sensor connector
21. Strip progress sensor connector
22. Unused strip bin sensor connector
23. BUS cable connector
26. Strip turner motor connector
29. Feeder motor connector
33. Strip pusher motor interface board



1. Strip feeder module

Components:

1. Test strip groove
2. Feeder blade
3. Flipped strip sensor
4. Strip turner ramp
5. Feeder drum's scoop



a,



b,



c,

Setting up the flipped strip sensor

- a, without strip:** Flipped strip sensor is inactive (green) and that the real-time value next to it is below the „Strip detect” parameter value (by default: 120)
- b, flipped strip:** the real-time value is above the „Strip turning” value (by default: 200)
- c, pads facing up:** Flipped strip sensor is active (red) and that the real-time value is between the „Strip detect” and „Strip turning” limits (by default, from 120 to 200).

Note:

The above mentioned values are recommendations but they can vary on each instrument and shall be adjusted if necessary.

The strips must be oriented properly so the dark blue ASC pad should be at the front.

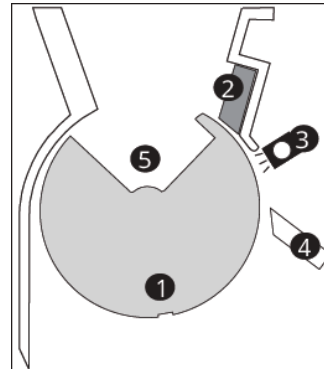
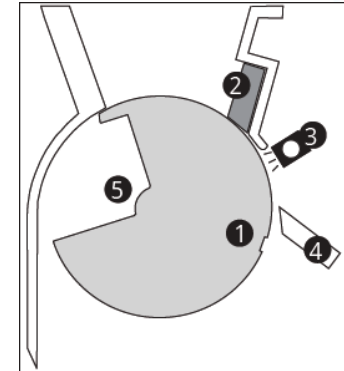
1. Strip feeder module

Feeder drum's positions:

- 5. Feeder motor „home“ position
- 8. Feeder motor „drop“ position
- 12. Feeder motor „upper empty“ position
- 16. Feeder motor „lower empty“ position

• drop position

The test strip groove (1) is moved clockwise (when seen from the front) below the level of the strip turner ramp (4).

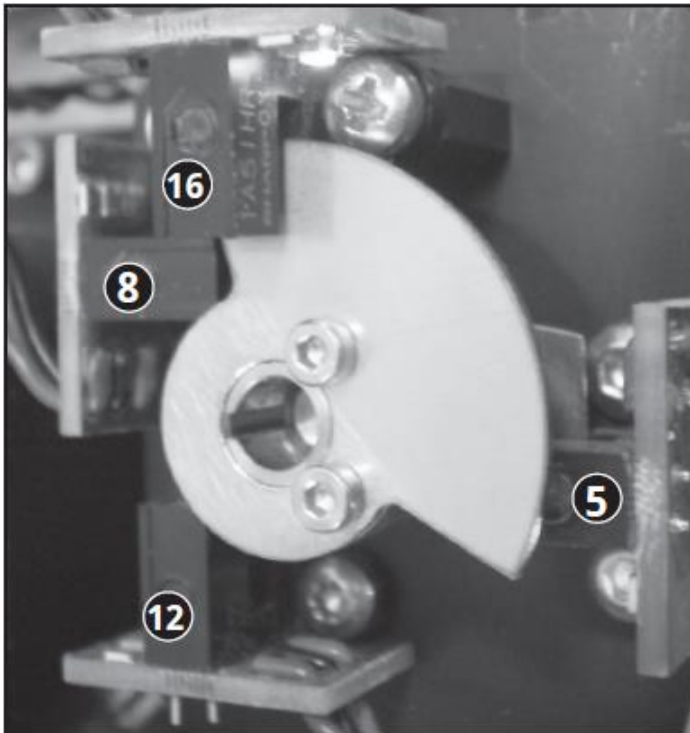
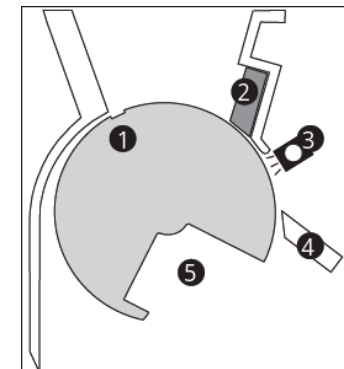


• upper empty

The feeder drum's scoop (5) is turned clockwise (when seen from the front) towards the feeding tube opening, ready to receive strips.

• lower empty section

The feeder drum's scoop is turned clockwise (when seen from the front) towards the unused strip bin, emptying itself. The test strip groove (1) is aligned with the edge of the feeder module bracket.

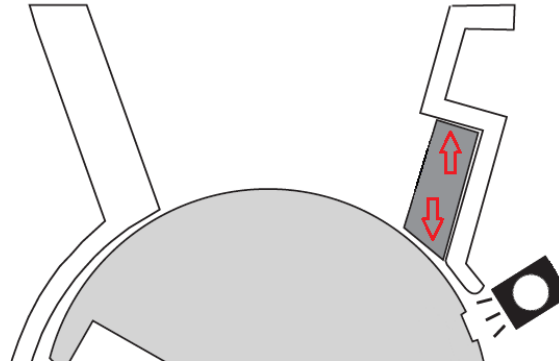




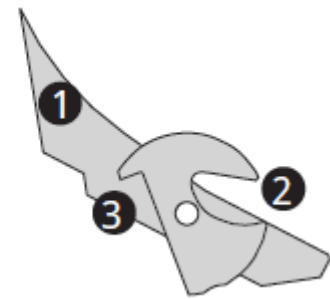
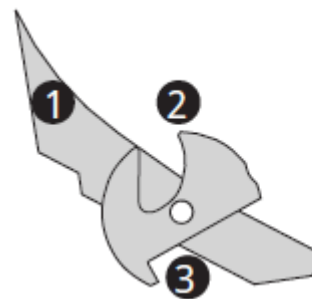
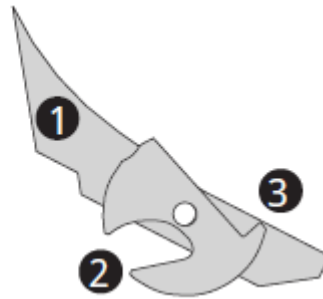
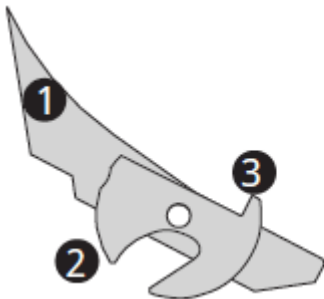
1. Strip feeder module

Feeder blade's position:

The clearance between the feeder drum and the feeder blade should be 0.15 millimeters so that only one test strip sitting in the feeder drum's groove can pass underneath the blade.



Strip turner's positions:

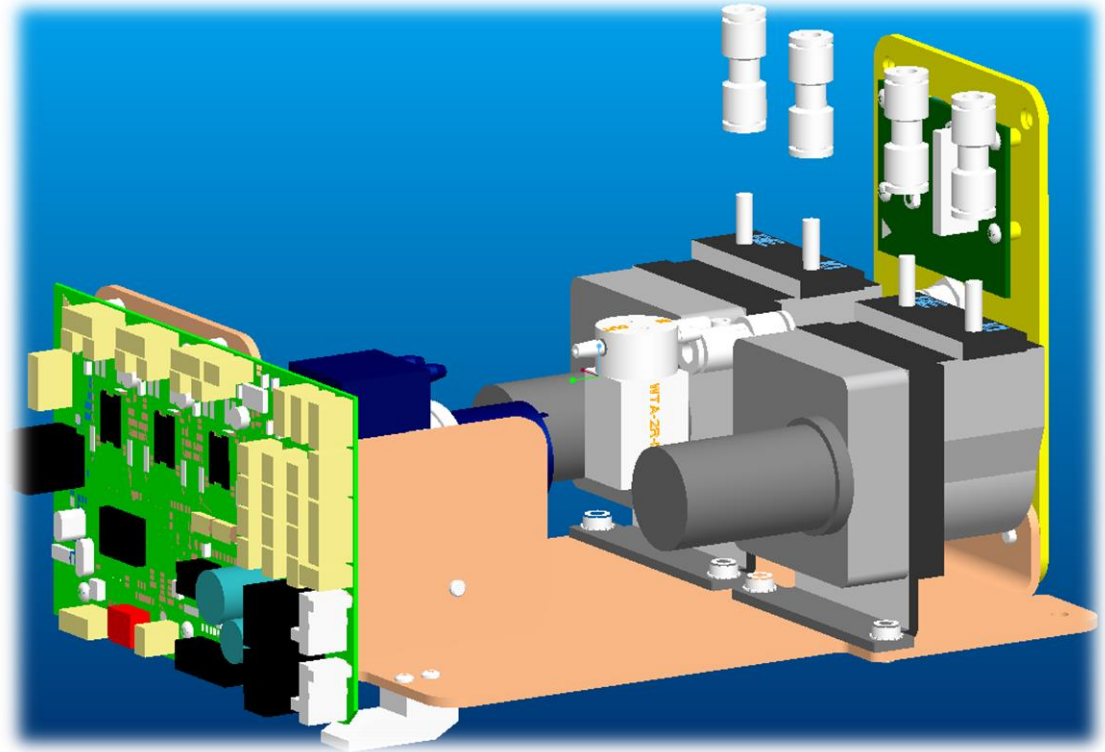




2. Washer module

Main components

- Inner washer peristaltic pump
- Outer washer peristaltic pump
- Membrane suction pump
- Waste valve



Module description:

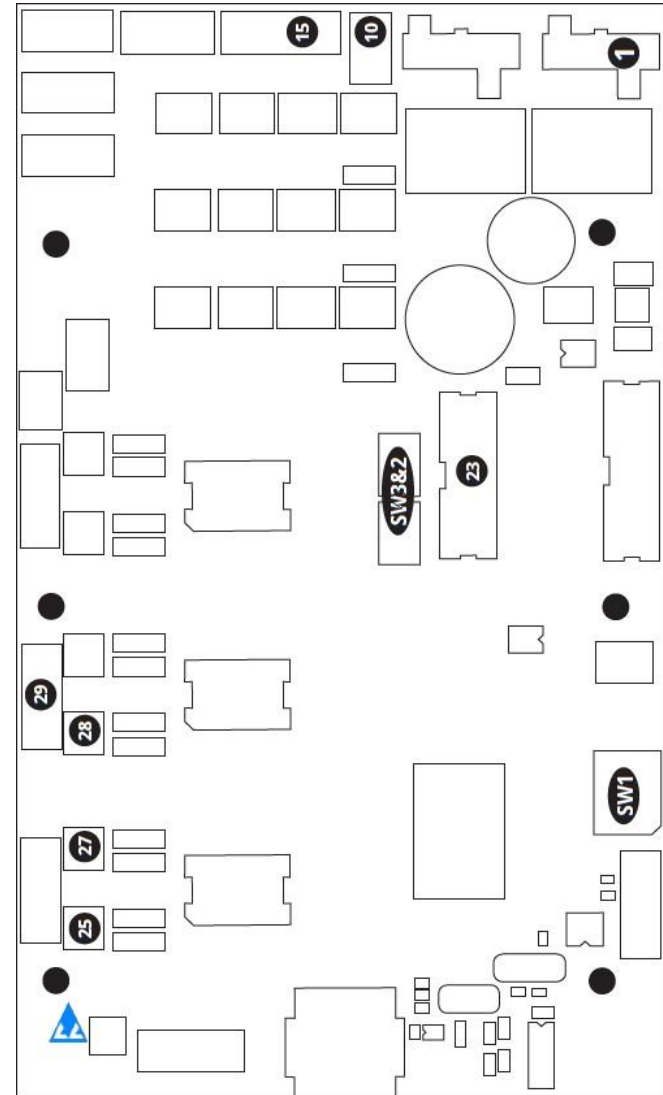
The Washer module is responsible for cleaning the inner and outer surface of the probe and sucking away the waste.



2. Washer module

PCB layout:

1. Power connector coming from the Rack mover PCB
10. Liquid tank sensor interface board connector
15. Liquid tank sensor interface board connector
23. BUS connector
25. Outer washer peristaltic pump motor connector
27. Inner washer peristaltic pump motor connector
28. Waste line sealing valve connector
29. Membrane pump motor connector

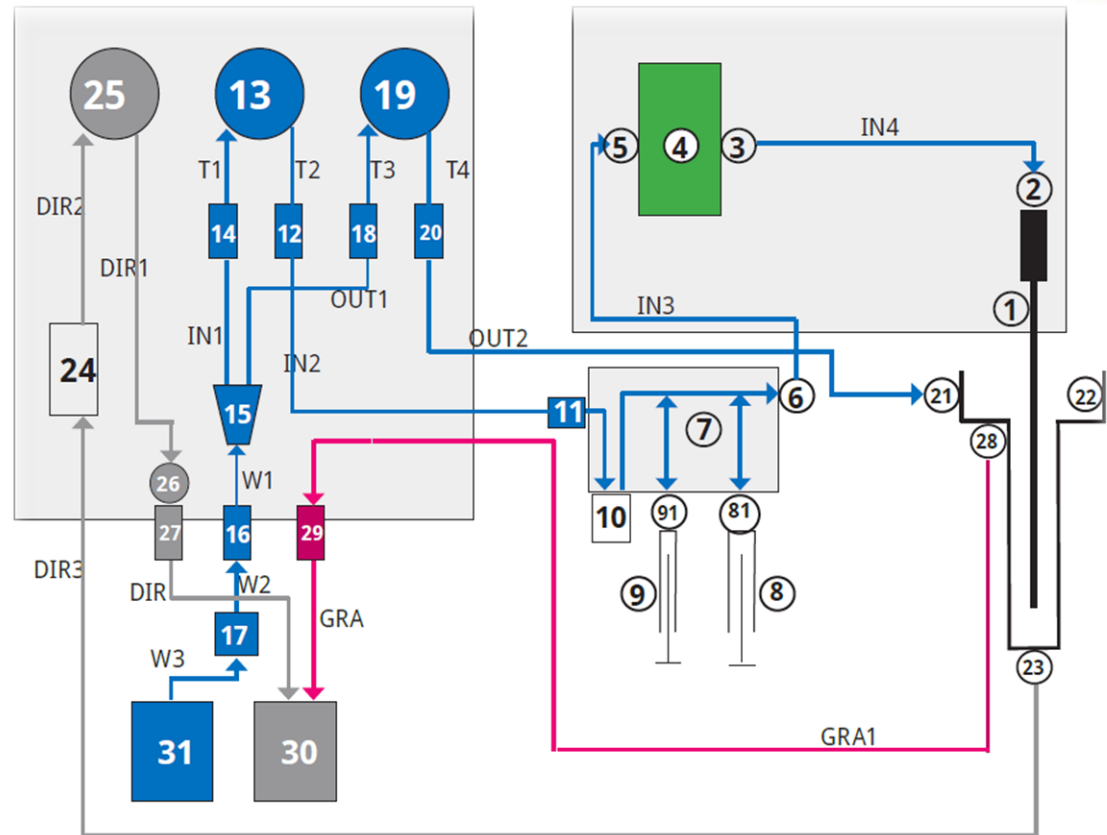
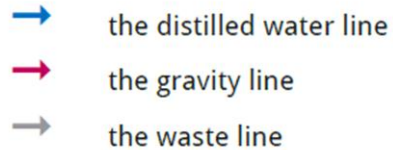




2. Washer module tubing diagram

Main components

1. Probe
4. PMC
7. Manifold
8. Hamilton syringe (2.5ml)
9. Hamilton syringe (0.25ml)
10. Sampler valve
13. Inner washer peristaltic pump
17. Filter
19. Outer washer peristaltic pump
22. Rinse station
24. Waste valve
25. Suction / membrane pump



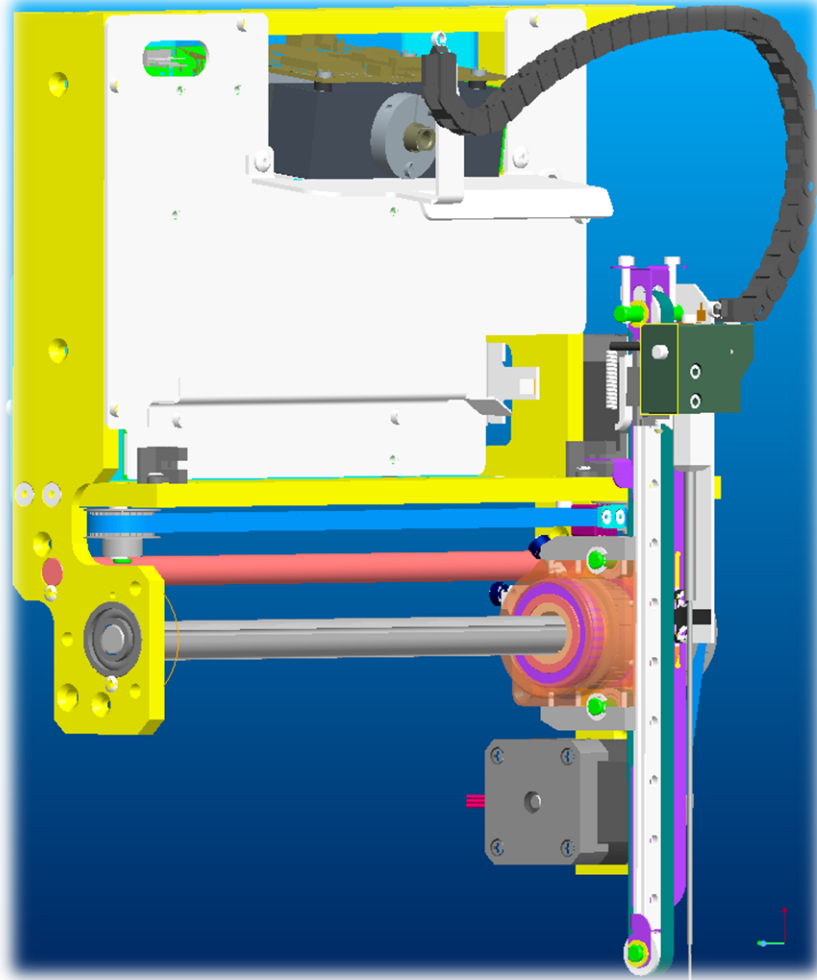
Yearly maintenance up to 300 samples per day:

- Waste line tubing (marked by gray)
- Hamilton syringes with O-rings
- Polyurethane tubing of the peristaltic pumps

3. Pipetting module

Main components

- Y (up-down) motor
- X (back-front) motor
- pipette
- PMC module



Module description:

The Pipetting module is responsible for the sample taking and dispensing it onto each pads.



3. Positioning the pipette above the pads (Position Setup)

Mechanical adjustments:

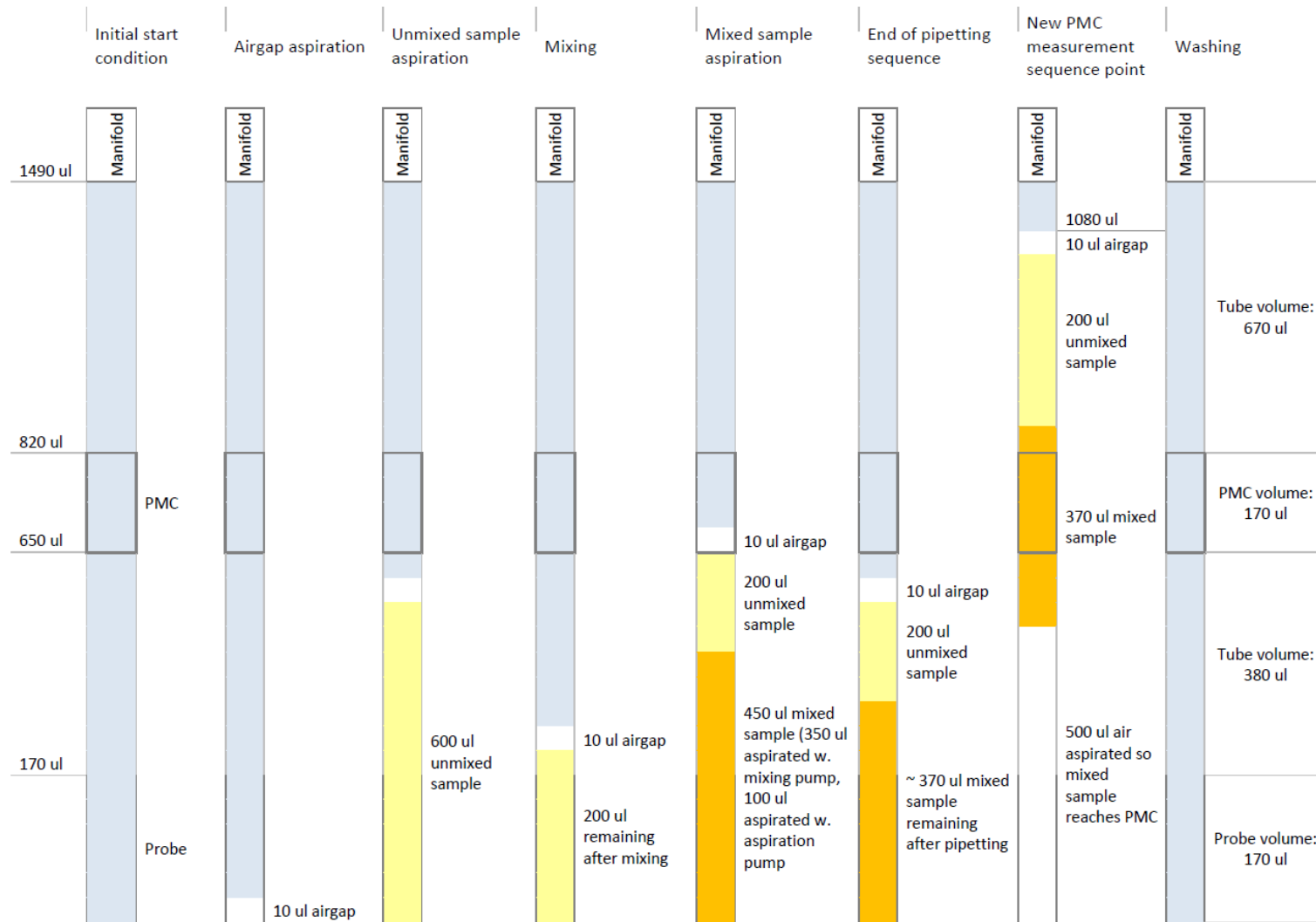
- Find the center of the tube with the help of the special positioning screw
- Check the parallelity of the pipette and the strip
- Finally move the rinse station to the center of the pipette.

Software adjustments:

- Navigate to position setup in the service software
- Take the pipette to the center of the pad
- Use a piece of paper to find the proper distance between the pipette and the pad



Sample mixing and aspirating sequence

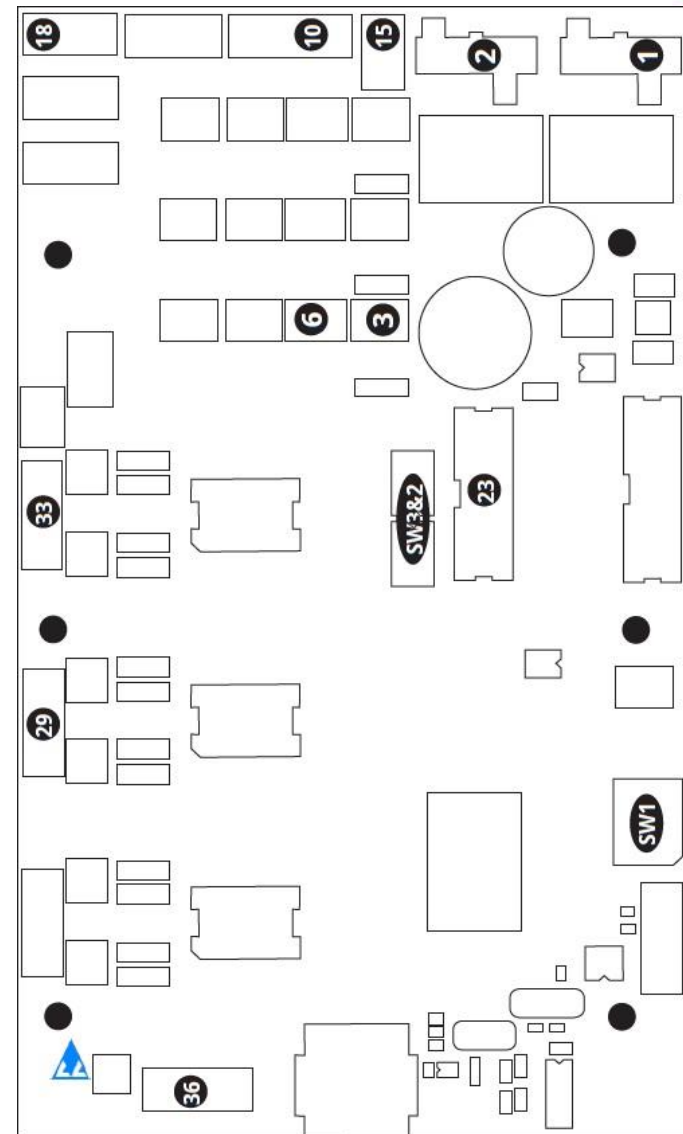




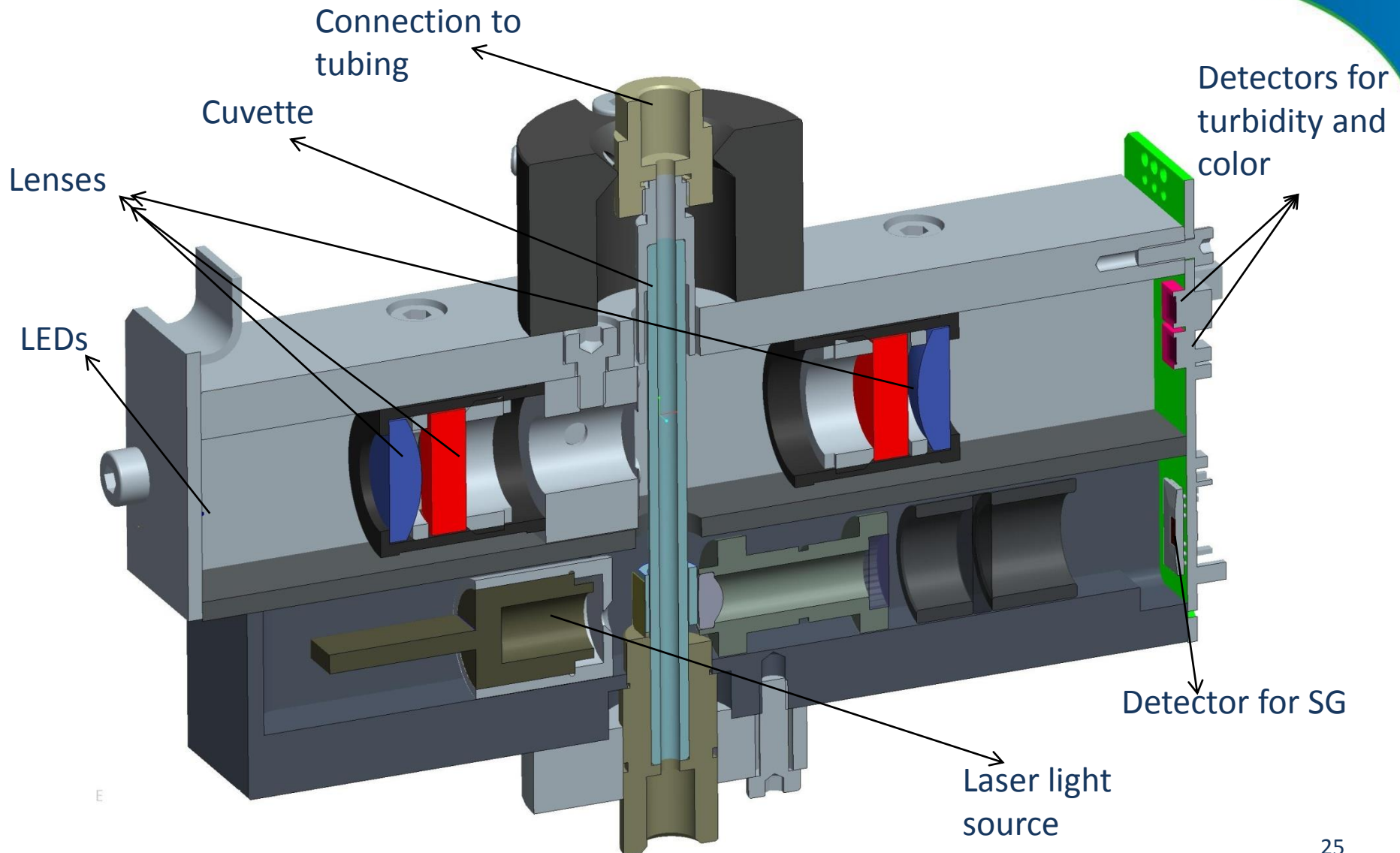
3. Pipetting module

PCB layout:

1. Power connector to the oscillator interface PCB
2. Power connector to Feeder PCB
3. Rinse well position sensor connector
6. Test tube position sensor connector
10. BUS connector
15. Right door inductive sensor connector
18. Left door inductive sensor connector
23. BUS cable connector
29. Linear robot vertical motor connector
33. Linear robot horizontal motor connector
36. PMC connector

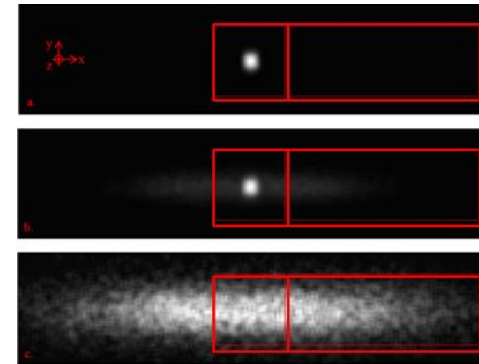
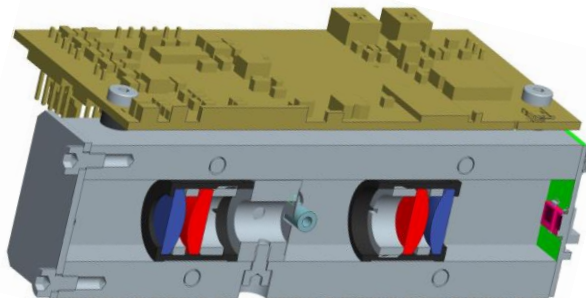


PMC - Physical Measurement Cell

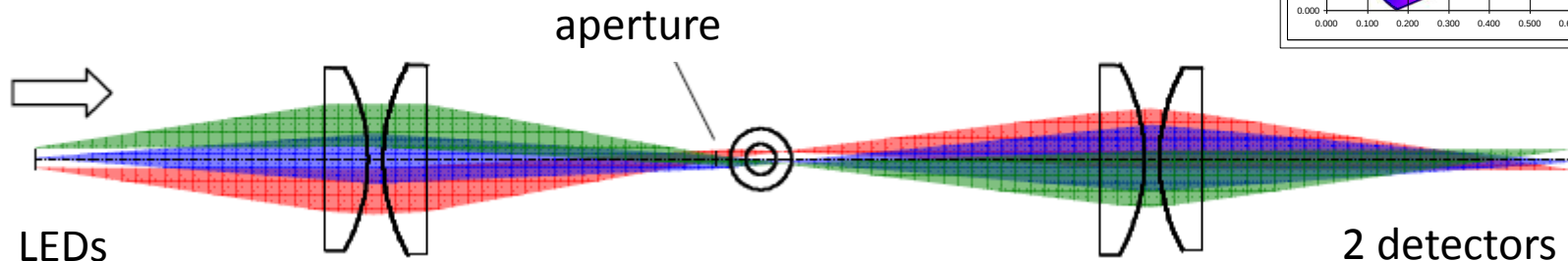
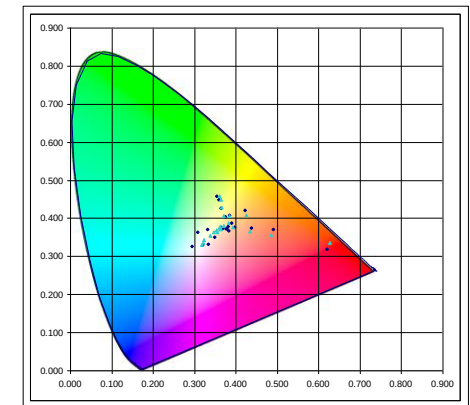


PMC – Turbidity & Color measurement

4 different colored LEDs (yellow, blue, green, red) light beams pass through the sample and the amount of direct and scattered light is measured.



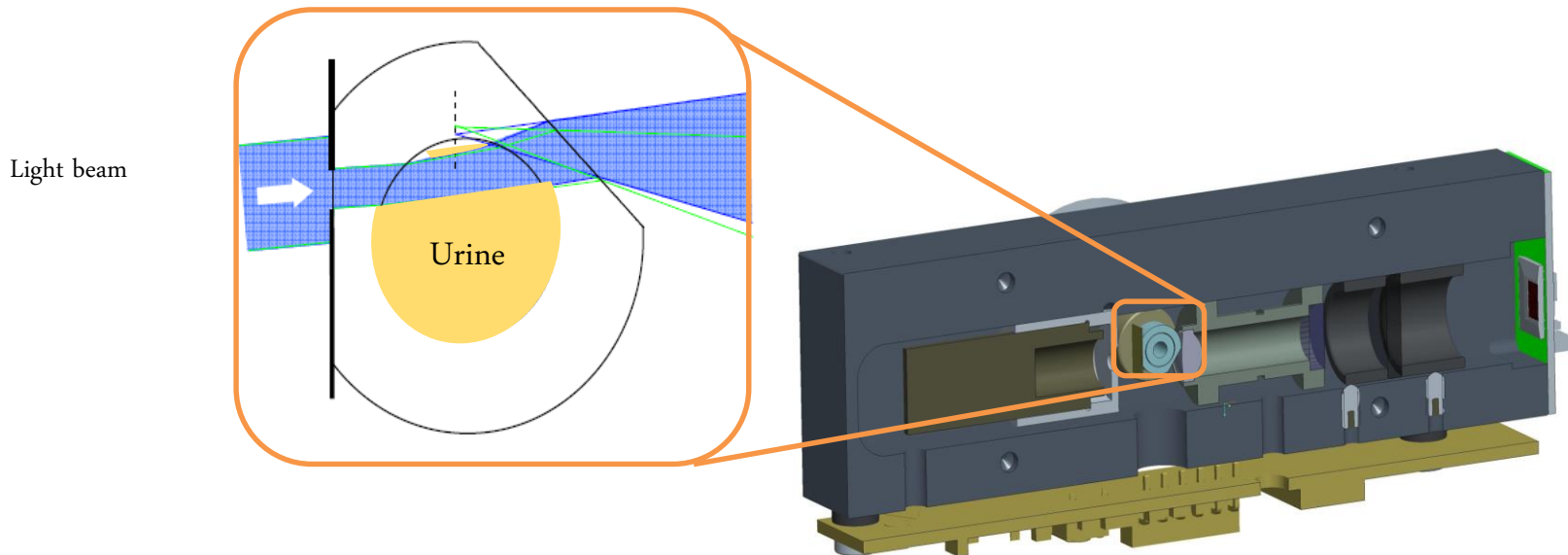
In case of color measurement the same optical arrangement is used as in the case of turbidity measurement. The theoretical background is the same as that of the turbidity measurement.



PMC – SG measurement



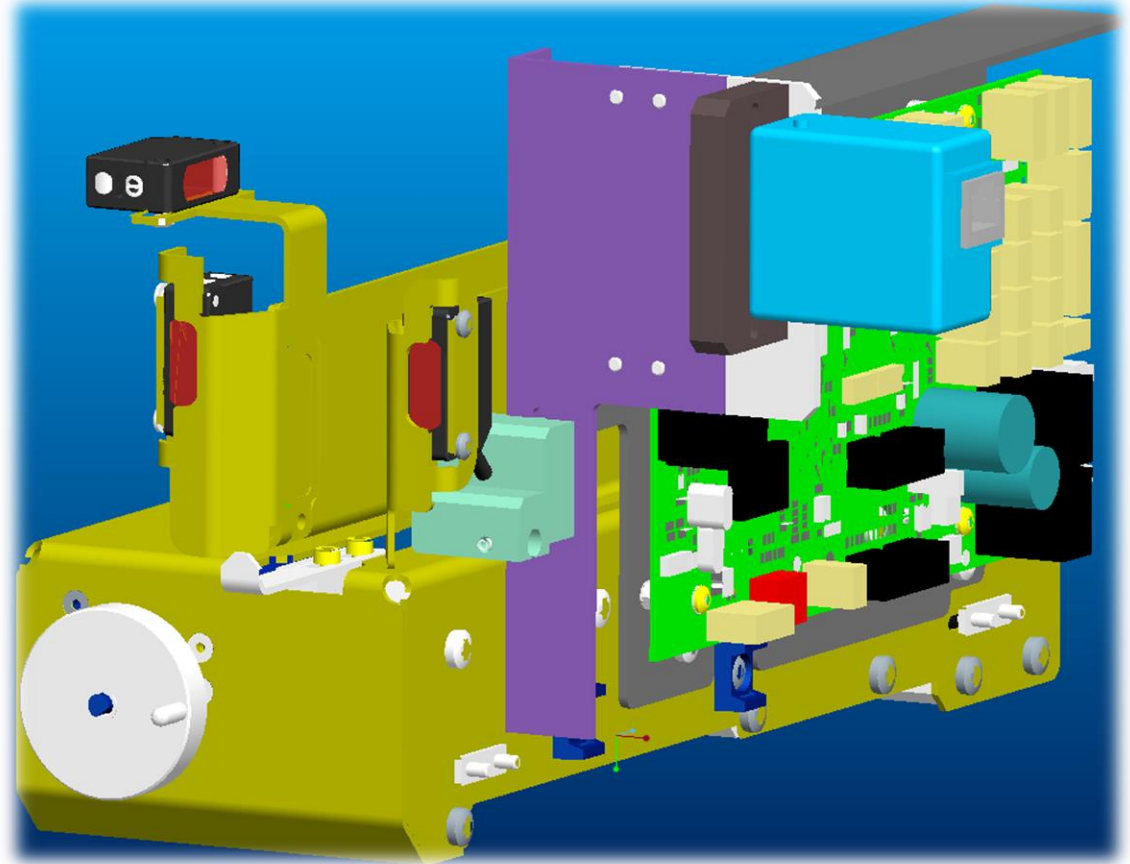
Based on the different refractive indexes of urine with various specific gravity
The light beam travels through the cuvette and is refracted twice on the borderline between the wall of the cuvette and the sample.
The level of refraction depends on the refractive index of the sample.



4. Rack module

Main components

- Rack aligner module
- Conveyor motor (DC motor)
- Rack mover motor
- Barcode reader
- Tube sensor
- Rack arrive and full sensors
- Sample level detector.



Module description:

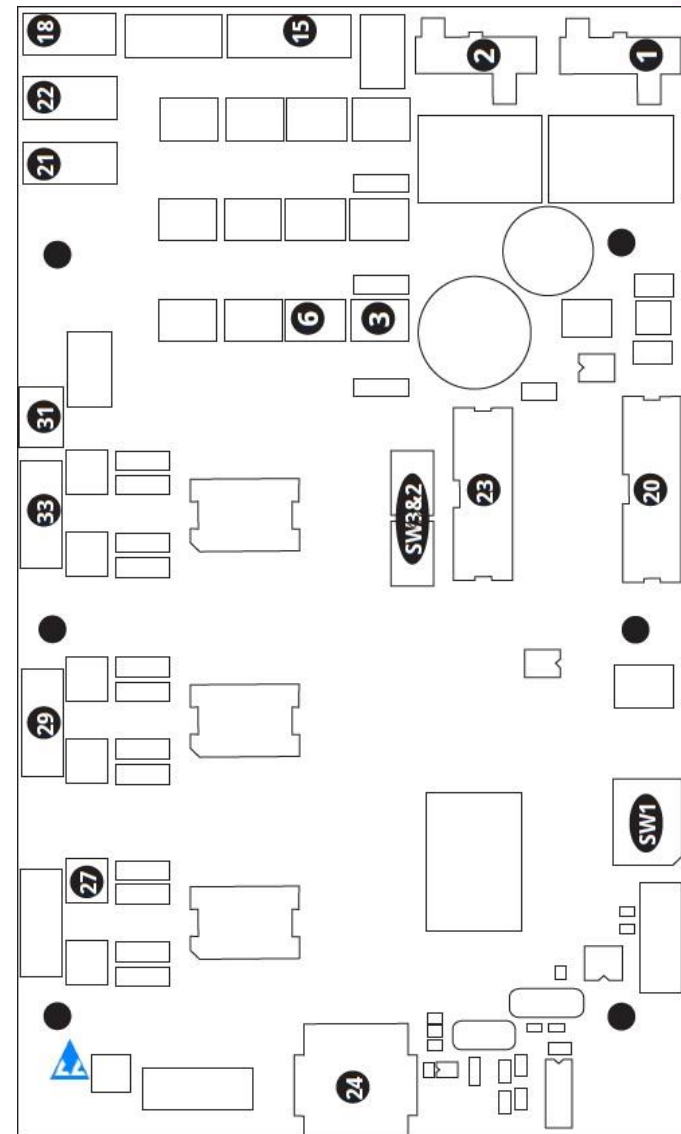
The Rack module is responsible for the rack transportation and positioning.



4. Rack module

PCB layout:

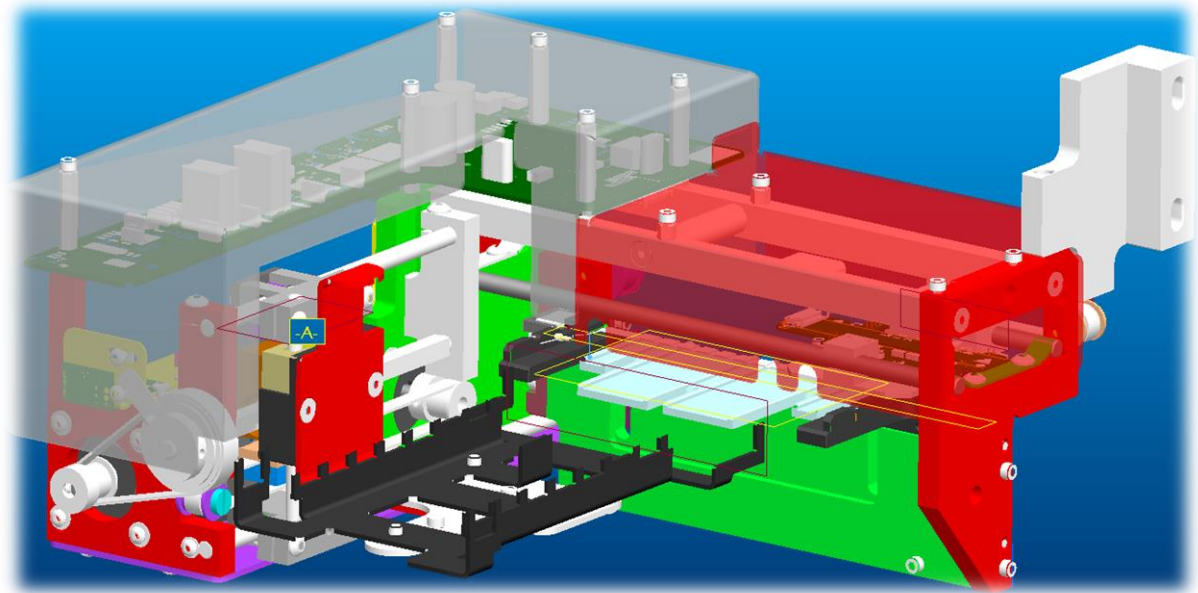
1. Power connector for the power supply
2. Power connector to the Washer PCB
3. Innermost catch position sensor connector
6. Outermost catch position sensor connector
15. Rack aligner sensor connector
18. Conveyor maximum capacity sensor
20. Level detector interface PCB connector
21. Tube presence sensor connector
22. Rack delivery sensor connector
23. BUS connector
24. Ethernet connector used by the barcode reader
27. Rack mover motor connector
29. Rack positioning motor connector
31. Rack positioning sensor connector
33. Rack aligner motor interface PCB connector



5. Strip timer and Photometer module

Main components

- Comb conveyor -> Must keep it clean otherwise the strips may fall down or give X flagged results.
- Photometer
- Strip hold down springs (left and right) -> yearly maintenance is needed!
- Grey reference pad -> yearly maintenance is needed!



Module description:

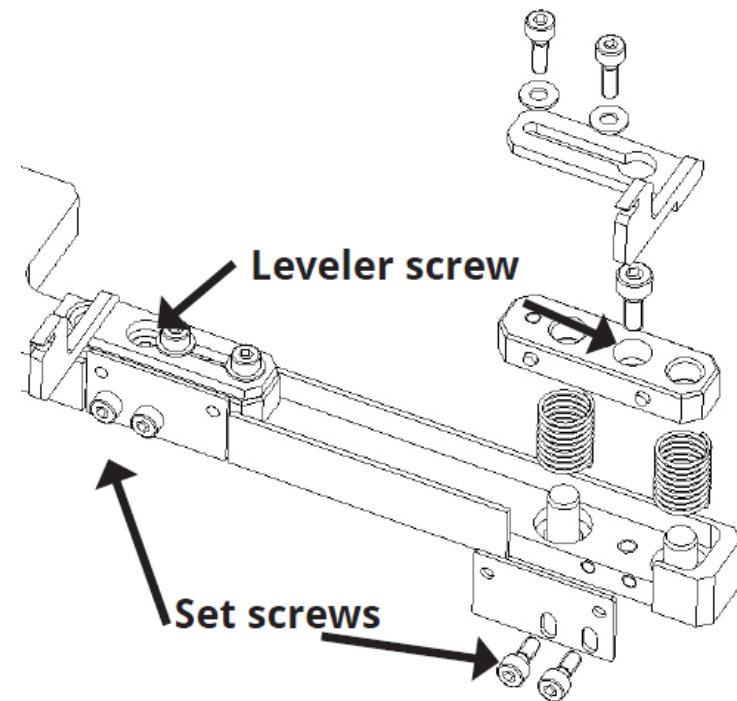
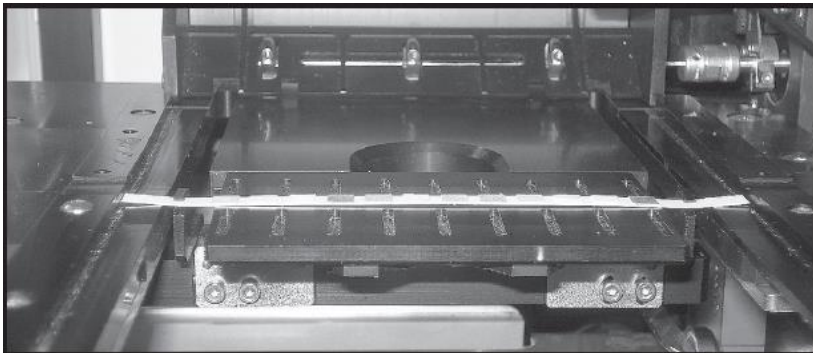
The Strip timer and Photometer module is responsible for forwarding and measuring the strips.



5. Adjusting the position of strip clamps

Adjustment:

- Move the pushing arm into pipetting position
- Remove the pipetting tray
- Lose the Set screws
- Adjust the level with the help of the leveler screws
- Tighten the Set screws back

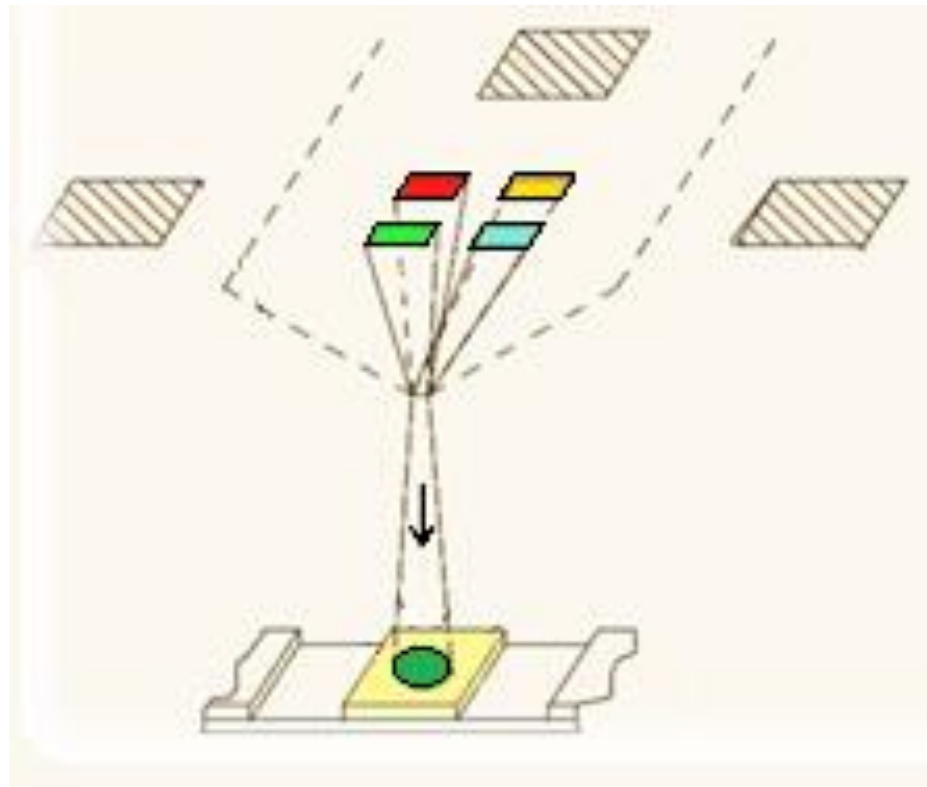


5. Positioning the strip under the measuring head (Position setup)



Measuring head position setup:

- Remove the right side cover
- Remove the cover of the measuring head
- Use the position setup to adjust the position of the light beam
- The green light-spot has to be in the center of the pads

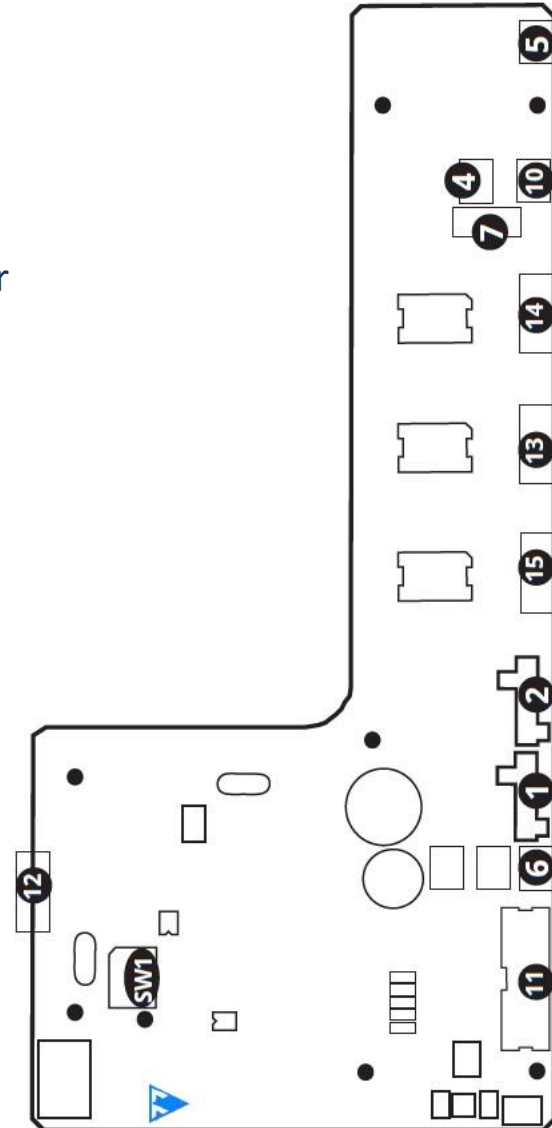




5. Strip timer and Photometer module

PCB layout:

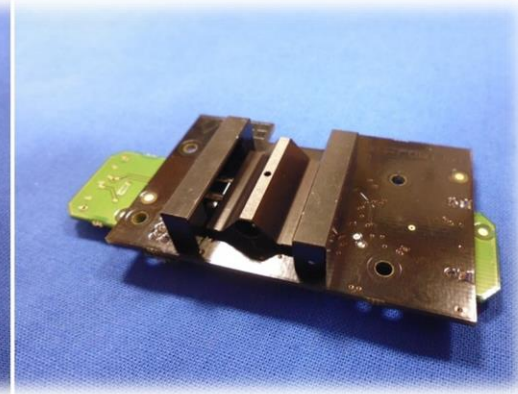
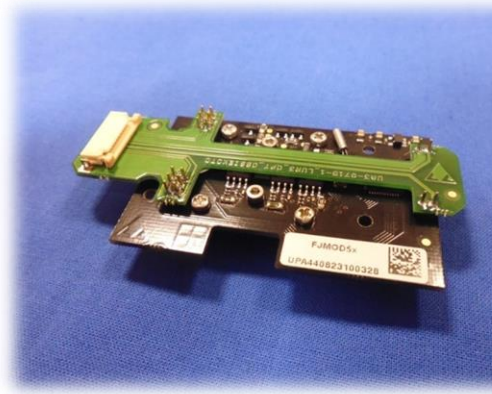
1. Power connector for the power supply
2. Power connector to the dispenser PCB
4. Horizontal strip timer home position sensor connector
5. Vertical strip timer motor home position sensor connector
6. Photometer home position sensor connector
7. Strip timer comb presence sensor connector
10. Strip bin micro-switch sensor connector
11. BUS connector
12. Photometer interface PCB
13. Horizontal strip timer motor connector
14. Vertical strip timer motor connector
15. Head mover motor connector





5. Strip timer and Photometer module

Photometer:



LabUMat 2: Automated test strip analysis

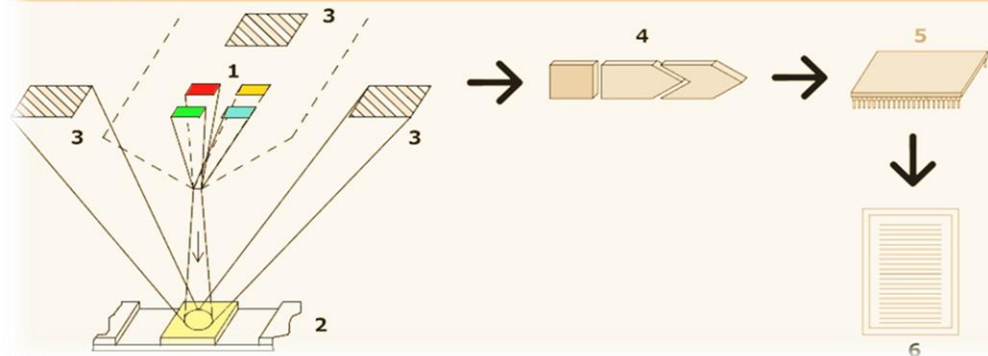
using reflectance photometry

LEDs shine light of defined wavelengths (1: 505, 530, 620, and 660 nm) on the test pad (2)

Detectors (3) pick up the reflected light

An A/D converter (4) changes the detector signal to digital form

The microprocessor (5) calculates a reflectance value based on a calibration standard, and compares them to range limits for a semiquantitative output (6)

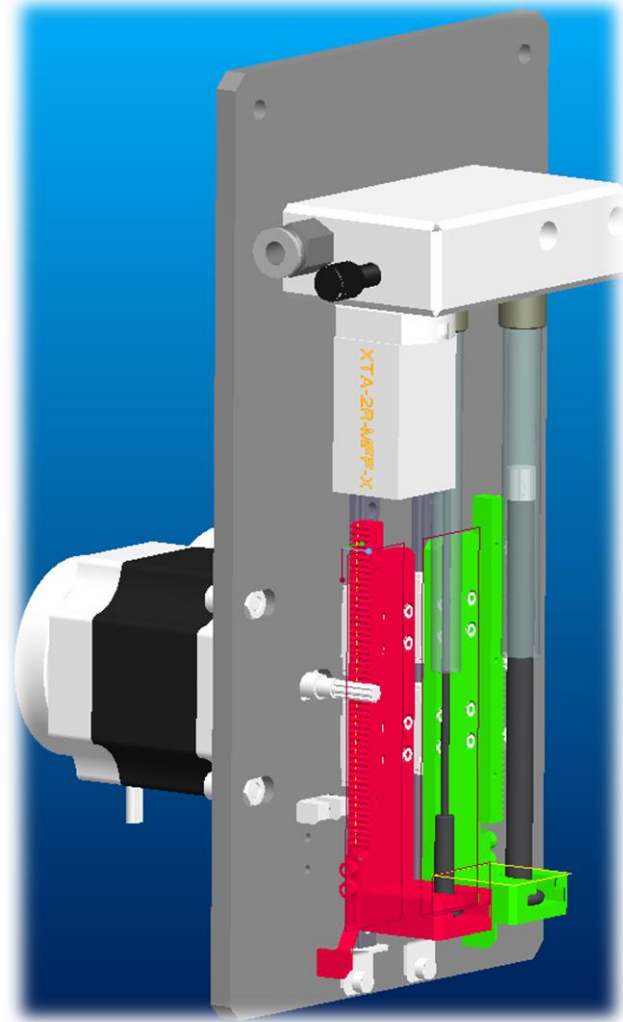




6. Sampler module

Main components

- Hamilton syringe 2.5 mL
- Hamilton syringe 250 μ L
- Valve

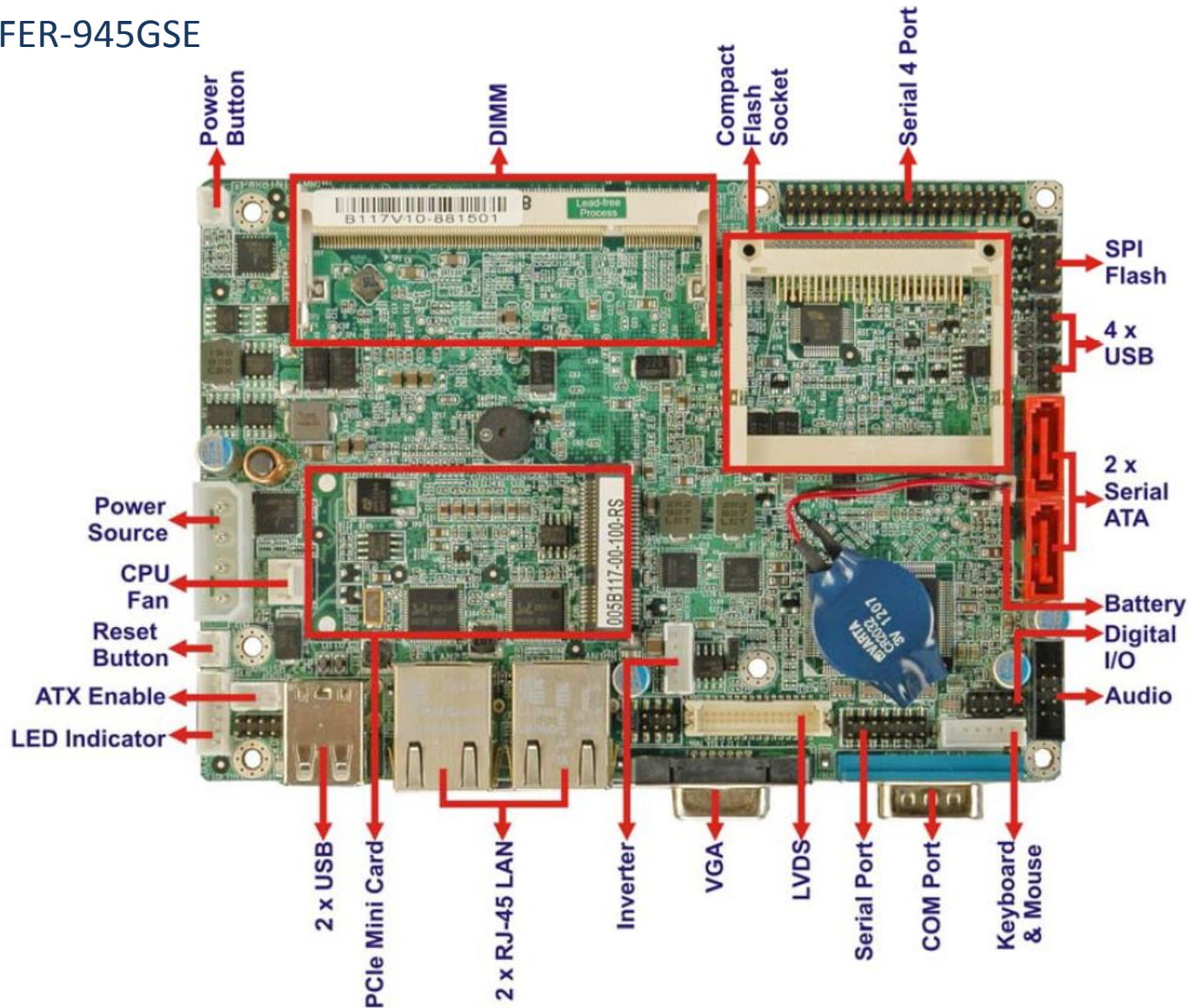


Module description:

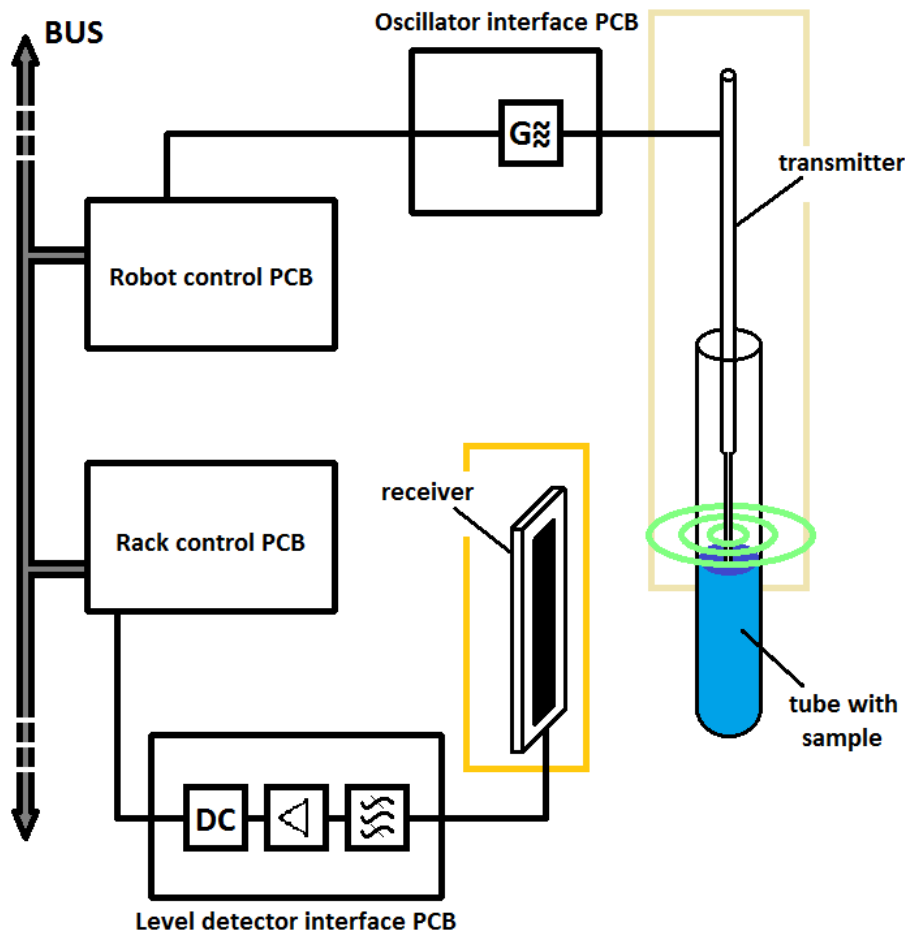
The Sampler module is responsible for the mixing and dispensing of the sample.

LabUMat 2 PC motherboard

WAFER-945GSE

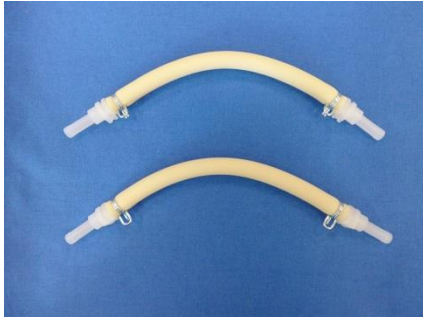


Level detection



- Based on an oscillator circuit; a 32 kHz signal is transmitted and the signal change is detected
- The probe transmits this 32 kHz signal
- A silver metal plate on the side of the rack module is the receiver
- Depending on the height position of the probe and the interference of the sample in the sample tube, a different signal is transmitted
- The position of the probe can be determined
- Sample volume: min. 2.0 ml
- Detected values:
 - Probe is in the sample ~550
 - Probe is out of the sample ~480

Service maintenance parts



Polyurethane
tubing for
peristaltic pumps



Filter



Hamilton syringes
with O-rings
250µl and 2.5 ml

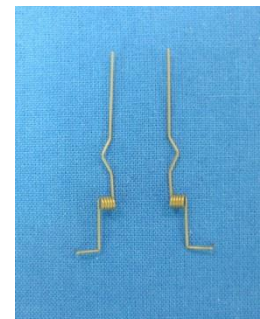


Reference pad



Pipes

GRA1+PU_ATM8/6
GRA1+PU_ATM6/4
DIR1+PU_ATM6/4
DIR2+PU_ATM6/4
DIR3+PU_ATM6/4



Strip-retaining springs